

Code Checklist	
Project Name	ToDdList-APP
Reviewed Document	Software Design Description
Reviewed Feature/Scope	Overall architecture, service design, UI design, database design, real-time communication, role-based permission, chatbot integration
Number of Pass	48
Number of Fail	11
Number of N/A	0
Number of Errors	

No.	Checklist Question	Review Result (Pass/Fail/N/A)			Note
		Pass	Fail	N/A	
1	Do the design specifications fulfill the main objectives of the system?	☒	☐	☐	
2	Does the design clearly map each major requirement/use case to corresponding modules or services?	☒	☐	☐	
3	Have all important system modules been included?	☐	☐	☐	
	a. Auth/User/Permission modules?	☒	☐	☐	
	b. Task/Folder/Note modules?	☒	☐	☐	
	c. Group/Notification/Chat modules?	☒	☐	☐	
	d. Admin and Amanda Chatbot modules?	☒	☐	☐	
4	Have common and shared functions been defined?	☐	☐	☐	
	a. Authentication and authorization?	☒	☐	☐	
	b. Input validation and error handling?	☒	☐	☐	
	c. Notification, logging, and audit functions?	☒	☐	☐	
5	Does the design follow a clear Client–Server architecture with RESTful API and Socket.IO real-time communication?	☒	☐	☐	
6	Are the interfaces between components clearly identified?	☐	☐	☐	
	a. Web/Mobile ↔ Backend API?	☒	☐	☐	
	b. Backend ↔ MongoDB?	☒	☐	☐	
	c. Backend ↔ SocketIO	☒	☐	☐	
	d. Frontend ↔ Amanda Chatbot API?	☒	☐	☐	
7	Are the interfaces consistent and easy to use in request/response format, naming, and expected behavior?	☐	☐	☐	
8	Are the modules sufficiently independent and self-contained to support maintainability?	☒	☐	☐	
9	Is the design flexible enough to add future features such as analytics, new business roles, or richer chatbot functions?	☐	☒	☐	
10	Have exception/error conditions been handled?	☐	☐	☐	
	a. Invalid input, missing data, or unauthorized access?	☒	☐	☐	
	b. Offline/connection issues?	☒	☐	☐	
	c. Notification, file upload, or chatbot service failures?	☐	☒	☐	File/notification errors are covered, but chatbot failure handling is not fully detailed.
11	Are error messages, warnings, and system feedback adequate?	☒	☐	☐	
12	Are security procedures included?	☐	☐	☐	
	a. Authentication and password protection?	☒	☐	☐	
	b. Role-based access control?	☒	☐	☐	
	c. Group/folder/task access restriction?	☒	☐	☐	
	d. Sensitive data protection?	☒	☐	☐	
13	Are audit/logging requirements included, especially admin action logs and login history?	☒	☐	☐	
14	Are the standard operating environments mentioned?	☐	☐	☐	
	a. User/Admin permissions?	☒	☐	☐	
	b. PO/PM/QA/BA/Developer roles?	☒	☐	☐	
	c. DevOps/Security/Cloud Infra/Chatbot roles?	☒	☐	☐	
15	Does the design prevent unauthorized users from viewing, editing, assigning, or deleting tasks outside their permission scope?	☒	☐	☐	

16	Does the software requirements specification include assumptions made?	☐	☐	☐	
	a. User, Task, Group, Folder, Note?	☒	☐	☐	
	b. Notification, Chat, File?	☒	☐	☐	
	c. Chatbot State, AdminActionLog, LoginHistory?	☒	☐	☐	
17	Are MongoDB collections designed appropriately?	☐	☐	☐	
	a. Fields and data types are suitable?	☒	☐	☐	
	b. References/embedded documents are reasonable?	☒	☐	☐	
	c. Validation rules are considered?	☒	☒	☐	
18	Are important query fields identified for indexing?	☐	☐	☐	
	a. userId, groupId, assigneeId?	☐	☒	☐	Not explicitly listed as index fields.
	b. task status, due date, priority?	☐	☒	☐	Search/filter exists, but index strategy is missing.
	c. notification receiver, login history userId, admin log targetId?	☐	☒	☐	Not explicitly specified.
19	Does the database design consider data integrity?	☐	☐	☐	
	a. Prevent orphan tasks/notes/files?	☐	☒	☐	Not clearly specified as database integrity rules.
	b. Validate group membership and assignee?	☒	☐	☐	
	c. Maintain consistent permission records?	☒	☐	☐	
20	Does the design address scalability and data volume for tasks, messages, notifications, files, and login logs?	☐	☐	☐	
21	Have performance bottlenecks been considered?	☐	☐	☐	
	a. Large task lists, search, and filtering?	☐	☒	☐	Search/filter is specified, but optimization strategy is not.
	b. Real-time chat and notification delivery?	☐	☒	☐	Real-time behavior is specified, but load/performance design is weak.
	c. Chatbot response time?	☐	☒	☐	Chatbot flow is described, but response-time bottleneck handling is not.
22	Are backup, restore, fallback, or recovery procedures mentioned for important user/task data?	☐	☒	☐	Backup/monitoring/cloud integration appears mainly as future direction, not current concrete design.
23	Are UI layouts and flows consistent with the specified use cases?	☐	☐	☐	
	a. Login/register/profile/settings screens?	☒	☐	☐	
	b. Task List/Kanban/Calendar views?	☒	☐	☐	
	c. Chat, Notification, and Admin Dashboard screens?	☒	☐	☐	
	d. Mobile screens?	☒	☐	☐	
24	Does the UI design include validation for important input fields?	☐	☐	☐	
	a. Email/password?	☒	☐	☐	
	b. Task title, due date, priority?	☒	☐	☐	
	c. Role, group member, message, and chatbot prompt?	☒	☐	☐	
25	Is navigation between screens clear and consistent across web and mobile interfaces?	☒	☐	☐	
26	Are real-time behaviors designed clearly?	☐	☐	☐	
	a. Task update synchronization?	☒	☐	☐	
	b. New message and notification delivery?	☒	☐	☐	
	c. Online/offline handling?	☒	☐	☐	
27	Is the Amanda Chatbot design connected clearly to input processing, context retrieval, model/training logic, and frontend interaction?	☒	☐	☐	
28	Are test plans or review evidence available to verify design correctness for core modules and high-risk flows?	☒	☐	☐	
Conclusion	Fully Approved	☐			
	Conditionally Approved	☒			
	Rejected	☐			
Discussion	The software design is acceptable for review because the report and SRS cover the main architecture, modules, use cases, UI flows, real-time behavior, permission logic, and chatbot design. The strongest points are the clear functional coverage, detailed business-role authorization, and broad SRS acceptance criteria. However, several design-level weaknesses remain, especially MongoDB schema detail, indexing strategy, API request/response standard, backup/restore procedure, scalability, and performance bottleneck analysis. These issues do not invalidate the design, but they should be recorded as improvement items before final approval.				